

SWATHI PADMANABHAN

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🌐 [Website](#)

Experience

TANUH (Translational AI for Networked Universal Healthcare) AICoE <i>Research Staff Member</i>	January, 2026 – Present <i>Bangalore, India</i>
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Education

Indian Institute of Science <i>PhD and Mtech (by Research), Department of Instrumentation and Applied Physics</i>	Aug. 2019 – July, 2025 <i>Bangalore, India</i>
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L.B.S Institute of Technology for Women <i>Bachelors of Technology in Applied Electronics and Instrumentation</i>	Aug. 2015 – June 2019 <i>Thiruvananthapuram, India</i>
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Publications

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1. **Swathi Padmanabhan**, Sarikonda Aryan Shashank, Rajesh Srinivasan and Jaya Prakash, “*Optoacousto-fluidics based optical rotation measure for blood glucose sensing*,” vol. 72, no. 11, pp. 3312-3323, Nov. 2025, doi: 10.1109/TBME.2025.3566940. [[Journal Article](#)]
 2. **Swathi Padmanabhan** and Jaya Prakash, “*Deep Tissue Sensing of Chiral Molecules using Polarization Enhanced Photoacoustics*,” Science Advances, 11, eado8012 (2025). DOI:10.1126/sciadv.ado8012. [[Journal Article](#)] [[News Coverage](#)]
 3. **Swathi Padmanabhan** and Jaya Prakash, “*Detecting optical rotary dispersion from photoacoustic circular dichroism of glucose in the NIR-II wavelength*”, IEEE Journal of Selected Topics in Quantum Electronics, doi: 10.1109/JSTQE.2025.3610710 [[Journal Article](#)]
 4. **Swathi Padmanabhan** and Jaya Prakash, “*Optimal Pathlength Identification for Accurate Glucose Sensing with Photoacoustic Derived Optical Rotation*,” Optics Letters, 50, 149-152 (2025). [[Journal Article](#)]
 5. Abhigyan Goswami, **Swathi Padmanabhan**, Sarthak Dash, Jaya Prakash and V.R. Supradeepa, “*Pulsed cascaded Raman fiber laser widely tunable in the second near-infrared and visible window for hyperspectral photoacoustic imaging*”, Optics Letters, 50, 2223-2226 (2025). [[Journal Article](#)][[Optica Spotlight](#)]
 6. **Swathi Padmanabhan***, Madhavi Tripathi*, Jaya Prakash and Ashok Raichur, “*Seed mediated galvanic synthesis of ultrasmall CuS-Au nanohybrids for photo-theranostic applications*,” ACS Applied Nano Materials 6, 14861-14875 (2023). [[Journal Article](#)] [[News Coverage](#)]
 7. Atheeth. S and Chandrashekar L N and Isha Munjal and **Swathi Padmanabhan** and Jaya Prakash and Manish Arora, “*Microfabrication of a 16 MHz 1D-pMUT-Array for photoacoustic and ultrasound imaging*,” Ultrasonic Imaging. 2025;0(0). doi:10.1177/01617346251380297 [[Journal Article](#)]

* denotes equal contribution

Patent

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1. **Swathi Padmanabhan** and Jaya Prakash, “*Apparatus for measuring concentrations of chiral molecules*” Indian Patent; Patent Number: IN547866 (2024). [[Granted](#)] [[Patent](#)]

Book Chapters

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1. Anandhu Anil Sindu, **Swathi Padmanabhan**, and Jaya Prakash. “Emerging Approaches in Metabolic Imaging Using Photoacoustics.” Bioimaging Modalities in Bioengineering, vol 9. Springer, Cham (2025): 529-545. [[Book Chapter](#)]
 2. **Swathi Padmanabhan** and Uma Ranjan, “*Markov Chain Monte Carlo*,” In: Daya Sagar, B.S., Cheng, Q., McKinley, J., Agterberg, F. (eds) Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer, Cham (2022). [[Book Chapter](#)]

Conferences and Workshops

- Abhigyan Goswami, **Swathi Padmanabhan**, Sarthak Dash, Jaya Prakash, V. R. Supradeepa, “Photoacoustic Spectroscopy Using a Widely Tunable Pulsed Cascaded Raman Fiber Laser,” 2025 Conference on Lasers and Electro-Optics Europe & European Quantum Electronics Conference (CLEO/Europe-EQEC), Munich, Germany, 2025, pp. 1-1. (23-27 June 2025) [[Conference Abstract](#)]
- Abhigyan Goswami, **Swathi Padmanabhan**, Sarthak Dash, Jaya Prakash, V. R. Supradeepa, “Passively Q-switched Raman fiber laser source widely tunable in near infrared and visible window,” Proc. SPIE 13347, Nonlinear Frequency Generation and Conversion: Materials and Devices XXIV, 133470T (21 March 2025). [[Conference Paper](#)]
- Abhigyan Goswami, **Swathi Padmanabhan**, Sarthak Dash, Jaya Prakash, V.R. Supradeepa, “Widely tunable Raman fiber laser for hyperspectral photoacoustic imaging,” Proc. SPIE 13319, Photons Plus Ultrasound: Imaging and Sensing 2025, 1331911 (20 March 2025). [[Conference Paper](#)]
- **Swathi Padmanabhan** and Jaya Prakash, “Approaching Optical Rotation Through the Lens of Near-Infra-Red-II Photoacoustics,” Optical Tomography and Spectroscopy, OTu3D. 6, Optica Biophotonics Congress: Biomedical Optics (2024). [[Conference Abstract](#)] [[Oral Presentation](#)]
- **Swathi Padmanabhan** and Jaya Prakash, “Exploring Photoacoustics from Optical Activity: A Perspective for Biomolecular Sensing,” 2024 IEEE South Asian Ultrasonics Symposium (SAUS) (2024). [[Oral Presentation](#)]
- **Swathi Padmanabhan** and Jaya Prakash, “Mathematical Modelling for Polarization Based Photoacoustic System to Evaluate the Changes in Propagation of Polarized Light in Tissues,” International Conference on Photonics, Photonics (2023). [[Conference Poster](#)]
- **Swathi Padmanabhan** and Jaya Prakash, “Mathematical Modelling for Polarization Based Photoacoustic System to Evaluate the Changes in Propagation of Polarized Light in Tissues,” Workshop on Medical Image Computing, Indian Institute of Science (2023). [[Workshop Poster](#)]
- **Swathi Padmanabhan** and Jaya Prakash, “Development of Polarisation based Photoacoustic Imaging System and Appropriate Mathematical models to enable the Propagation of Polarized Light in Tissues”, Workshop on The role of AI in Transforming Healthcare, Indian Institute of Science (2022). [[Workshop Poster](#)]

Relevant Coursework

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| <ul style="list-style-type: none">• Biomedical Optics and Spectroscopy• Medical Imaging• Optical Metrology• Digital Image Processing• Advanced Image Processing | <ul style="list-style-type: none">• Numerical Methods• Sensors and Transducers• Biology for Engineers• Drug Delivery |
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Technical & Experimental Expertise

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| <ul style="list-style-type: none">• Microfluidic Channel Design & Fabrication (Soft Lithography)• CAD designing for 3D printing• Instrumentation - optomechanics, system calibration, motion actuator control• Ultrasound system response characterisation• Optics - Laser Safety, Optical alignment, Polarization optics, Laser energy calibration• Operation of Pulsed lasers, diode lasers, CW lasers and Raman Fiber lasers• Electronics -Data acquisition with Arduino, Raspberry Pi & TI AFE5818EVM evaluation boards. | <ul style="list-style-type: none">• Cell Culture (Beginner’s level) - HeLa Cells• Operation of UV-VIS Spectrophotometer• Operation of Oxygen Plasma for bonding techniques in fabrication• Operation of US single element, linear array and PMUT transducers for US data acquisition• Devising Human Ethics protocols• Patient studies and involvement with doctors and clinicians• Signal conditioning for post-processing of data• Mathematical formulations for acoustic signal analysis for optical concepts• Ultrasound and optical Phantom preparation |
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- Post-processing of experimental Ultrasound data
- Handling Patient data and samples
- *Ex-vivo* experimental design for therapy and sensing
- *In-vivo* experimental optimisation
- Implementation of Polarised Monte Carlo with depthwise optical activity

Technical Skills

Languages: C, MATLAB, Python, C++

Developer Tools: Visual Studio Code, LabVIEW

Academic Service/Outreach

Reviewer: IEEE Transactions on Medical Imaging, IEEE Transactions on Circuits and Systems I, Journal of Biomedical Optics, IEEE Photonics, Physica Scripta

Teaching: Teaching Assistance for courses Biomedical Optics and Spectroscopy, Medical Imaging

Outreach: Institute (IISc) Open Day demos, Outreach with high school students on prospective higher education

Professional Societies: Student Member of SPIE and Optica. Student Member of Applied Photonics Initiative (API) at IISc.